Course: Linear Algebra **Instructor:** Sajal Rana
Assignment #1

Problem 1.

(a) Let

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix}.$$

Find A^2 and A^3.

Solution. First, we compute $A^2 = A \cdot A$:

$$\begin{aligned} A^2 &= \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} (0)(0)+(0)(2)+(1)(1) & (0)(0)+(0)(1)+(1)(-2) & (0)(1)+(0)(0)+(1)(0) \\ (2)(0)+(1)(2)+(0)(1) & (2)(0)+(1)(1)+(0)(-2) & (2)(1)+(1)(0)+(0)(0) \\ (1)(0)+(-2)(2)+(0)(1) & (1)(0)+(-2)(1)+(0)(-2) & (1)(1)+(-2)(0)+(0)(0) \end{bmatrix} \\ &= \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 2 \\ -4 & -2 & 1 \end{bmatrix}. \end{aligned}$$

Next, we compute $A^3 = A^2 \cdot A$:

$$\begin{aligned} A^3 &= \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 2 \\ -4 & -2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} (1)(0)+(-2)(2)+(0)(1) & (1)(0)+(-2)(1)+(0)(-2) & (1)(1)+(-2)(0)+(0)(0) \\ (2)(0)+(1)(2)+(2)(1) & (2)(0)+(1)(1)+(2)(-2) & (2)(1)+(1)(0)+(2)(0) \\ (-4)(0)+(-2)(2)+(1)(1) & (-4)(0)+(-2)(1)+(1)(-2) & (-4)(1)+(-2)(0)+(1)(0) \end{bmatrix} \\ &= \begin{bmatrix} -4 & -2 & 1 \\ 4 & -3 & 2 \\ -3 & -4 & -4 \end{bmatrix}. \end{aligned}$$

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(b) Show that $A^3 = A^2 + A - 5I$.

Solution. We calculate the right-hand side $A^2 + A - 5I$:

$$\begin{aligned} A^2 + A - 5I &= \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 2 \\ -4 & -2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1+0-5 & -2+0-0 & 0+1-0 \\ 2+2-0 & 1+1-5 & 2+0-0 \\ -4+1-0 & -2-2-0 & 1+0-5 \end{bmatrix} \\ &= \begin{bmatrix} -4 & -2 & 1 \\ 4 & -3 & 2 \\ -3 & -4 & -4 \end{bmatrix}. \end{aligned}$$

Since this equals A^3, we have verified that:

$$\boxed{A^3 = A^2 + A - 5I}.$$

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(c) Using (b), without doing any more multiplication, prove that:
(i) $A^4 = 2A^2 - 4A - 5I$
Solution. Starting with $A^4 = A \cdot A^3$, and using $A^3 = A^2 + A - 5I$ from part (b):

$$\begin{aligned} A^4 &= A(A^2 + A - 5I) \\ &= A^3 + A^2 - 5A \end{aligned}$$

Substituting $A^3 = A^2 + A - 5I$ into the equation again:

$$\begin{aligned} A^4 &= (A^2 + A - 5I) + A^2 - 5A \\ &= 2A^2 - 4A - 5I. \end{aligned}$$

Hence proved.

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(ii) $A^{-1} = \frac{1}{5}(I + A - A^2)$
Solution. From part (b), we have:

$$A^3 = A^2 + A - 5I.$$

Rearranging the terms:

$$5I = A^2 + A - A^3.$$

Dividing by 5:

$$I = \frac{1}{5}(A^2 + A - A^3).$$

Multiplying A^{-1} on both sides:

$$IA^{-1} = \frac{1}{5}(A^2A^{-1} + AA^{-1} - A^3A^{-1}).$$

Using the properties $A^2A^{-1} = A$, $AA^{-1} = I$, and $A^3A^{-1} = A^2$:

$$A^{-1} = \frac{1}{5}(A + I - A^2) = \frac{1}{5}(I + A - A^2).$$

Hence proved.

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Problem 2.

Show that the most general matrix that commutes with

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

is of the form

$$\begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}.$$

Solution. Let X be a general 3×3 matrix:

$$X = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}.$$

The condition that X commutes with M means $XM = MX$.
First, we compute the product XM:

$$XM = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a & b \\ 0 & d & e \\ 0 & g & h \end{bmatrix}.$$

Next, we compute the product MX:

$$MX = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} d & e & f \\ g & h & i \\ 0 & 0 & 0 \end{bmatrix}.$$

Equating XM and MX:

$$\begin{bmatrix} 0 & a & b \\ 0 & d & e \\ 0 & g & h \end{bmatrix} = \begin{bmatrix} d & e & f \\ g & h & i \\ 0 & 0 & 0 \end{bmatrix}.$$

By comparing corresponding entries row by row:

- From Row 1: $d = 0$, $e = a$, $f = b$.
- From Row 2: $g = 0$, $h = d$, $i = e$. Since $d = 0$ and $e = a$, we get $h = 0$ and $i = a$.
- From Row 3: $0 = 0$, $0 = 0$, $0 = 0$.

Substituting these relationships ($d = 0$, $g = 0$, $h = 0$, $e = a$, $i = a$, $f = b$) back into X:

$$X = \begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}.$$

Thus, the most general commuting matrix is indeed:

$$\boxed{\begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}}.$$

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Problem 3.

(a) Let A be a square matrix. Prove that $A + A^T$ is symmetric and $A - A^T$ is skew-symmetric.

Solution. Let $B = A + A^T$. Taking the transpose,

$$\begin{aligned} B^T &= (A + A^T)^T \\ &= A^T + (A^T)^T \\ &= A^T + A \\ &= A + A^T \\ &= B. \end{aligned}$$

Therefore, $B^T = B$, so $A + A^T$ is symmetric.
Now let $C = A - A^T$. Taking the transpose,

$$\begin{aligned} C^T &= (A - A^T)^T \\ &= A^T - (A^T)^T \\ &= A^T - A \\ &= -(A - A^T) \\ &= -C. \end{aligned}$$

Therefore, $C^T = -C$, so $A - A^T$ is skew-symmetric.

$$\boxed{A + A^T \text{ is symmetric and } A - A^T \text{ is skew-symmetric.}}$$

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(b) If A is an $m \times n$ matrix, prove that AA^T and A^TA are both symmetric.
Solution. Taking the transpose of AA^T,

$$\begin{aligned} (AA^T)^T &= (A^T)^T A^T \\ &= AA^T. \end{aligned}$$

Therefore, AA^T is symmetric.
Similarly,

$$\begin{aligned} (A^TA)^T &= A^T(A^T)^T \\ &= A^TA. \end{aligned}$$

Therefore, A^TA is symmetric.
Hence, both AA^T and A^TA are symmetric. ■

(c) If $A^2 = A$ and A^{-1} exists, prove that $A = I$.
Solution. Given $A^2 = A$. Since A^{-1} exists, multiply both sides by A^{-1}:

$$A^{-1}A^2 = A^{-1}A.$$

Using associativity,

$$(A^{-1}A)A = A^{-1}A.$$

Since $A^{-1}A = I$, we obtain

$$IA = I.$$

Therefore,

$$\boxed{A = I}.$$

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(d) If A is invertible, prove that $(A^{-1})^T = (A^T)^{-1}$.

Solution. Since A is invertible, there exists a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I.$$

Take the transpose of both sides of the equation $AA^{-1} = I$:

$$(AA^{-1})^T = I^T.$$

Using the properties of transpose,

$$(AA^{-1})^T = (A^{-1})^T A^T$$

and

$$I^T = I.$$

Thus,

$$(A^{-1})^T A^T = I.$$

Similarly, taking the transpose of $A^{-1}A = I$ gives

$$A^T (A^{-1})^T = I.$$

Therefore, $(A^{-1})^T$ is both a left and right inverse of A^T.

By uniqueness of the inverse, we conclude that

$$\boxed{(A^T)^{-1} = (A^{-1})^T}.$$

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Problem 4.

If A is a square matrix of order 3 such that $A^T = -A$, prove that the diagonal entries of A all equal 0.

Solution. Let

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}.$$

Since

$$A^T = -A,$$

the transpose of A is

$$A^T = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix},$$

while

$$-A = \begin{bmatrix} -a & -b & -c \\ -d & -e & -f \\ -g & -h & -i \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} = \begin{bmatrix} -a & -b & -c \\ -d & -e & -f \\ -g & -h & -i \end{bmatrix}.$$

Comparing the diagonal entries on both sides gives

$$a = -a, \qquad e = -e, \qquad i = -i.$$

Thus,

$$2a = 0, \qquad 2e = 0, \qquad 2i = 0.$$

Hence,

$$a = 0, \qquad e = 0, \qquad i = 0.$$

Therefore, all the diagonal entries of A are zero:

$$\boxed{a = e = i = 0}.$$

Hence, every skew-symmetric matrix of order 3 has zero diagonal entries. ■

Problem 5.

Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, and $X = \begin{bmatrix} x & y \\ z & t \end{bmatrix}$. We are given the equation $AX = XB$.

Solution. First, we compute the product AX using the definition of matrix multiplication :

$$AX = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x & y \\ z & t \end{bmatrix} = \begin{bmatrix} x+0 & y+0 \\ x+2z & y+2t \end{bmatrix} = \begin{bmatrix} x & y \\ x+2z & y+2t \end{bmatrix}$$

Next, we compute the product XB :

$$XB = \begin{bmatrix} x & y \\ z & t \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 2x-y & -x+2y \\ 2z-t & -z+2t \end{bmatrix}$$

Since two matrices are equal if and only if their corresponding entries are equal , equating AX and XB yields the following system of equations:

$$x = 2x - y \tag{1}$$
$$y = -x + 2y \tag{2}$$
$$x + 2z = 2z - t \tag{3}$$
$$y + 2t = -z + 2t \tag{4}$$

Simplifying each equation:

- From equation (1): $x - 2x = -y \implies -x = -y \implies x = y$
- From equation (2): $y - 2y = -x \implies -y = -x \implies x = y$
- From equation (3): Subtracting $2z$ from both sides gives $x = -t \implies t = -x$

- From equation (4): Subtracting $2t$ from both sides gives $y = -z \implies z = -y$. Since we already know $y = x$, substituting this gives $z = -x$.

Substituting $y = x$, $z = -x$, and $t = -x$ back into the matrix X:

$$X = \begin{bmatrix} x & y \\ z & t \end{bmatrix} = \begin{bmatrix} x & x \\ -x & -x \end{bmatrix}$$

By factoring out the scalar x from the matrix , we get:

$$X = x \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

Thus, X is a scalar multiple of $\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$. ■

Problem 6.

Let A and B be general 2×2 matrices with entries:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

Solution. First, we compute the product AB using standard matrix multiplication:

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$

Next, we find the transpose of this product, $(AB)^T$, by interchanging its rows and columns:

$$(AB)^T = \begin{bmatrix} ae + bg & ce + dg \\ af + bh & cf + dh \end{bmatrix}$$

Now, we compute the transposes of A and B individually:

$$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}, \quad B^T = \begin{bmatrix} e & g \\ f & h \end{bmatrix}$$

Then, we calculate the product $B^T A^T$:

$$B^T A^T = \begin{bmatrix} e & g \\ f & h \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} ea + gb & ec + gd \\ fa + hb & fc + hd \end{bmatrix}$$

Since multiplication of real numbers (scalars) is commutative (e.g., $ea = ae$ and $gb = bg$), we can rearrange the terms in the entries of $B^T A^T$:

$$B^T A^T = \begin{bmatrix} ae + bg & ce + dg \\ af + bh & cf + dh \end{bmatrix}$$

Comparing the resulting matrices, we can clearly see that both expressions yield the exact same matrix:

$$(AB)^T = B^T A^T$$

Thus, the result is verified for general 2×2 matrices. ■